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SYSTEM AND METHOD FOR PROVIDING ENERGY TO A CARBON CAPTURE INSTALLATION

Abstract

A system may include a CO₂ capture module, including a capture unit for capturing CO₂ from a gas using a capture material and a regeneration unit for unloading the CO₂ from a loaded capture material and regenerating said loaded capture material using heat. A system may include an energy module for producing electrical power and heat using solar energy, wherein at least a portion of the heat produced in the energy module is used in the regeneration unit.

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Background/Summary

PRIORITY [0001] The present application claims priority to and the benefit of U.S. Provisional Patent Application No. 63/562,719 filed Mar. 8, 2024, entitled “SYSTEM AND METHOD FOR PROVIDING ENERGY TO A CARBON CAPTURE INSTALLATION”, the disclosure of which is incorporated herein by reference in its entirety.

BACKGROUND

[0002] Carbon dioxide (CO₂) capture materials, such as carbon capture solvents, amines, and sorbent materials (metal-organic frameworks, Mesoporous carbon, etc.) require heat at a range of temperatures generally above 90° C., in particular between 105° C. and 150° C., to desorb CO₂ from the loaded capture material and regenerate said capture material so that it can capture additional CO₂ once regenerated.

SUMMARY

[0003] The disclosure relates to a system for CO₂ capture using a capture material with improved energy efficiency. The disclosure discloses an installation including a CO₂ capture module, including a capture unit for capturing CO₂ from a gas using a capture material and a regeneration unit for unloading the CO₂ from the loaded capture material and regenerating said capture material using heat and an energy module for producing electrical power and heat using solar energy, wherein the heat produced in the energy module is used in the regeneration unit.

[0004] The gas may be atmospheric air (i.e., direct air capture), a post-combustion flue gas, a natural gas, and/or any type of gas containing CO₂. The capture medium may be a solvent (such as an amine solvent, aqueous or non-aqueous), a sorbent (such as a mesoporous carbon, metal-organic framework, alkalized alumina), and/or an ion exchange material. The capture unit may include, for instance, an absorption column, a packed bed, and/or a rotating packed bed depending on the type of capture material.

[0005] The energy module includes a solar receiver that converts solar energy into electricity, for instance, a polycrystalline photovoltaic (PV) panel using high efficiency multijunction cells, and/or captures heat being produced by infrared and UV-A/UV-B wavelength of the solar spectrum, which is conventionally dissipated, for later use in a thermal storage. In some examples, the solar receiver both converts solar energy into electricity via a PV panel and captures heat via a fluid circulated therein (which may further cool the PV panel). The heatload is, for instance, stored at or below 100° C. in a hot thermal storage.

[0006] The electricity generated directly of the solar receiver may be used to run a compression chiller cycle. The compression chiller cycle may store cold thermal energy (i.e., coolth) in chilled fluid (in a cold storage) at a high coefficient of performance. In some examples, the cold thermal storage stores a liquid or slurry medium slightly above its freezing point. The electricity may also be used to power directly the CO₂ capture module (such as powering fluidic devices from the fluid circuit such as the circulation pumps, valves, impurity removal devices such as gas polishing units, etc.)

[0007] The available temperature differential between the hot thermal storage and cold thermal storage can be used to evaporate and condense a refrigerant, generating electricity via a thermodynamic cycle, such as an Organic Rankine Cycle (ORC) or Kalina cycle, and/or the hot or cold energy can be dispatched directly to the CO₂ module.

[0008] In the case of the CO₂ capture, byproduct solar heat near the boiling point of water becomes available and can provide a significant amount of the thermal energy required to desorb CO₂ from the CO₂ capture materials during regeneration.

[0009] In some examples, the capture material does not require a heat higher than the boiling point of water and the heat from the thermal storage may be used directly, without any power. In such

examples, the heat from the hot thermal storage may be used directly via passive heat exchange. In the case where the required regeneration temperature of the capture medium is greater than the boiling temperature of water (e.g., 105-125° C.), the hot fluid stored in the hot thermal storage may be further heated before being directed to the regeneration unit. The required heat may be obtained from the hot fluid coming from the energy module via a heat pump, and/or a heat exchanger (direct or indirect), with or without the use of a heat storing installation such as refractory bricks. In some examples, an additional heating unit includes a heat exchanger and/or heat pump that uses another source of heat (such as the flue gas heat, etc.), a resistive heating element, or a combination thereof. [0010] In some embodiments, the hot fluid is water and may be used directly (and/or after additional heating) in the regeneration unit to desorb the CO₂ from the loaded capture material. In another embodiment, the hot fluid exchanges heat with water circulating in a closed loop to deliver the required heat into the desorber material (optionally by vaporizing said water).

[0011] The electricity produced by the energy module may be used to provide the power required to run one or more of the heating units.

[0012] A significant amount of synergy value can be provided coupling the available heat from the energy module to the regeneration process. The use of renewable energy in the carbon capture installation enables to capture CO₂ from any source using less energy and further limiting the CO₂ indirectly needed for powering the CO₂ installation. Furthermore, having heat and electricity being available to the CO₂ module via the energy module enables to provide the regeneration heat in different manners, and being able to provide the heat regardless of the received solar ray, for instance in the middle of the night, as will be explained below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] In order to describe the manner in which the above-recited and other features of the disclosure can be obtained, a more particular description will be rendered by reference to specific embodiments thereof which are illustrated in the appended drawings. For better understanding, the like elements have been designated by like reference numbers throughout the various accompanying figures. While some of the drawings may be schematic or exaggerated representations of concepts, non-schematic drawings should be considered as being to scale for some embodiments of the present disclosure, but not to scale for other embodiments contemplated herein. Understanding that the drawings depict some example embodiments, the embodiments will be described and explained with additional specificity and detail through the use of the accompanying drawings in which:

[0014] FIG. 1 is a system diagram of an energy module including a solar harvesting system and a thermal cycle generator, according to at least some embodiments of the present disclosure.

[0015] FIG. 2 is a schematic diagram of a carbon dioxide (CO₂) capture installation coupling an energy module and a CO₂ module, according to at least some embodiments of the present disclosure.

[0016] FIG. 3 is a system diagram of another CO₂ capture installation, according to at least some embodiments of the present disclosure.

[0017] FIG. 4 is a system diagram of a CO₂ capture installation having a non-continuous (i.e., batch) adsorption-desorption process, according to at least some embodiments of the present disclosure.

[0018] FIG. 5 is a system diagram of an installation including an energy module with heat-storing solid units, according to at least some embodiments of the present disclosure.

[0019] FIG. 6 is a system diagram of another energy module of a CO₂ capture installation, according to at least some embodiments of the present disclosure.

[0020] FIG. 7 is a system diagram of a CO₂ module using high-temperature fluid of FIG. 6, according to at least some embodiments of the present disclosure.

[0021] FIG. 8 is another system diagram of a CO₂ module using high-temperature fluid of FIG. 6, according to at least some embodiments of the present disclosure.

[0022] FIG. 9 is a flowchart illustrating a method of capturing CO₂, according to at least some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0023] In the following, reference is made to embodiments of the disclosure. It should be understood; however, that the disclosure is not limited to specific described embodiments. Instead, any combination of the following features and elements, whether related to different embodiments or not, is contemplated to implement and practice the disclosure. Furthermore, although embodiments of the disclosure may achieve advantages over other possible solutions and/or over the prior art, whether or not a particular advantage is achieved by a given embodiment is not limiting of the disclosure. Thus, the following aspects, features, embodiments and advantages are merely illustrative and are not considered elements or limitations of the claims except where explicitly recited in a claim. Likewise, reference to “the disclosure” shall not be construed as a generalization of inventive subject matter disclosed herein and should not be considered to be an element or limitation of the claims except where explicitly recited in a claim.

[0024] Although the terms first, second, third, etc., may be used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms may be only used to distinguish one element, components, region, layer or section from another region, layer or section. Terms such as “first”, “second” and other numerical terms, when used herein, do not imply a sequence or order unless clearly indicated by the context. Thus, a first element, component, region, layer or section discussed herein could be termed a second element, component, region, layer or section without departing from the teachings of the example embodiments.

[0025] FIG. 1 illustrates such an example of an energy module **100** including a solar harvesting system and a thermal cycle generator **102**, for instance a generator using an ORC (Organic Ranking Cycle) or a Kalina cycle. An example of such solar energy harvesting system is provided below. One or more mirrors **104** direct sunlight **106** onto a solar thermal collector, such as raised photovoltaic (PV) modules **108** supported by a PV module tower **110**, that are actively cooled by water or other fluid circulated through the PV module tower **110**. The PV modules **108** convert the sunlight **106** to energy with approximately 90% efficiency, with about 30% of the sunlight energy converted to electricity by the PV modules **108** and about 60% converted to heat. The PV module may include a multijunction cell.

[0026] The heat is captured by a circulating fluid stream, and the heat is stored in a nearby water reservoir that is the hot energy storage (HES) reservoir (first storage reservoir) **112** or heat source for the thermal cycle generator **102**. The HES may contain the circulating water stream, or another fluid heated using the water stream. The fluid stored in the HES will in the following be designated as “hot fluid”.

[0027] The PV electricity may be used to power all or part of the CO₂ installation, for instance an additional heating unit as will be discussed later and/or fluid distribution devices (fluid circulation devices, valves, etc.). Optionally, the PV electricity (or the grid) may be used to power a refrigeration system **114** or chiller to cool water in a second reservoir that is the cold energy storage (CES) **116** (second storage reservoir), additionally and/or alternatively from powering the CO₂ installation. The streams from the HES and CES may be used as part of a thermal cycle generator to generate electricity. In particular, the thermal cycle generator **102** may be in thermal and hydraulic communication with an HES **112** and a CES **116** to provide the temperature differential to the thermal cycle generator. It uses elements to evaporate and/or condense a working fluid and a turbine for generating electricity. In some embodiments, the thermal cycle generator **102** is

configured to generate electrical power based on a temperature differential between the HES **112** and the CES **116**. In some embodiments without a CES **116**, the thermal cycle generator **102** is configured to generate electrical power based on a temperature differential between the HES **112** and an ambient heat sink, such as exhausting heat to the ambient atmosphere.

[0028] Excess electricity produced by the thermal cycle generator may be used to power a load from the CO₂ installations and/or sold to power local systems and/or sold to a power grid **118**. The two thermally insulated storage reservoirs, HES **112** and CES **116**, may maintain a temperature difference of approximately 90° C. For example, the temperature difference is approximately equivalent to the efficiency of a pumped hydrological system with a height difference of one kilometer. This temperature difference is exemplary, but other temperature differences may be maintained especially if the primary use of the CES is not to power the thermal cycle generator **102**.

[0029] FIG. **2** shows a schematic diagram of a carbon capture installation **200** coupling an energy module (not represented in its entirety) and a CO₂ module **201** according to some embodiments of the disclosure. In some embodiments, the installation **200** uses a solvent. For instance, an aqueous or non-aqueous solvent having an amine active compound.

[0030] Installation **200** includes a CO₂ module **201** including a capture unit **202** having one or more absorption stage **204** in series (in the shown example, three absorption stages **204**), a flue gas inlet **206** at the bottom of the capture unit, and a lean solvent inlet **208** at the top of the absorption stages **204** so that the lean solvent and the flue gas flow in the capture unit in counter-current (e.g., the lean solvent flows downward and the flue gas flows upward). Each absorption stage **204** may be configured to maximize contact between the flue gas and the lean solvent. The capture unit **202** includes an outlet **210** for the loaded (CO₂ rich) solvent at the bottom of the column. The cleaned flue gas exits the absorption stages **204** at the top of the absorption stages **204**. The capture unit **202** may include additional post-treatment units such as washing stages and impurity removal stage.

[0031] The CO₂ module **201** may also optionally include a cooling unit **212** to cool the flue gas before it enters the capture unit **202**. The CO₂ module **201** also includes a regeneration unit **214** including a regeneration column **216**, having one or more regeneration stages **218** in series, (in the shown example, three regeneration stages), a CO₂ rich solvent inlet **220** at the top of the column and a steam inlet **222** at the bottom of the column so that the rich solvent and the steam flow in counter-current. The regeneration unit also includes an outlet **224** for CO₂ loaded gas (mixed with steam) and an outlet for lean solvent **226**. The regeneration unit also includes a condenser **228** downstream of the outlet **224** for condensing the steam exiting the column and redirecting it to the column and a reclaimer **230** at the bottom of the regeneration column for further purifying the lean solvent (or at least a portion thereof). In some embodiments, the reclaimer **230** is a boiler that boils a portion of the lean solvent exiting from the column using steam provided via line **232** to generate a gas phase that is withdrawn from the reclaimer and returned in the regeneration column via inlet **222**, as well as a steam condensate that is withdrawn from the reclaimer at a temperature below boiling point (e.g., less than 90° C.) via line **234**. In some embodiments, the reclaimer **230** is a boiler that boils a portion of the lean solvent exiting from the column using any hot fluid having a temperature greater than 90° C. In some embodiments, the reclaimer **230** boils a portion of the lean solvent using a hot fluid having a temperature between 90° C. and 150° C. Lean absorbent exiting from the reclaimer **230** is redirected to the capture unit **202** via line **236**. Additional heat optimization may be provided within the CO₂ module itself (for instance, the heat exchanger between lean and rich solvent—see **238**).

[0032] The CO₂ module **201** that has been described in relationship to FIG. **2** is an exemplary module and could, in other embodiments, include different variants and additional elements. The energy module includes a hot energy storage pit (HES) **250**, as discussed in relationship with FIG. **1**. The fluid contained in the HES **250** is at approximately 95° C. therefore not sufficient to provide steam. The installation comprises a heat exchanger **252** for exchanging heat between the hot fluid

and the condensate that exits the reclaimers **230**, raising the temperature of the condensate to approximately 90° C. The hot fluid from the HES **250** is drawn from the top of the HES **250** and eventually reinjected into the HES **250**, generally at the bottom of the HES **250**, at a lower temperature, such as approximately 75° C. The fluid in the HES **250** may contain dissolved elements that assist in separating the fluid having a lower temperature from fluid having higher temperature using density to maintain a thermocline therebetween.

[0033] In some embodiments, the installation **200** includes a heat pump **254** for exchanging heat between the hot fluid stored in the HES **250** and the condensate that exits the heat exchanger **252**. As known in the art, the heat pump **254** includes a working fluid circuit using evaporation, compression, and condensation. The working fluid evaporates using the heat from the HES **250** (generally in a heat exchanger named evaporator) and is then compressed, increasing its temperature. Once compressed, it is cooled and condensed via heat exchange with the condensate (generally in a heat exchanger named condenser), enabling a further rise in the temperature of the condensate to approximately 120° C. The heat that may be used as an input to the heat pump may be hot fluid drawn from the top of the HES **250** at 95° C. and/or from the bottom of the HES **250** at 75° C. In some embodiments, in view of the intermediate working fluid circuit, the fluid from the HES **250** need not be hotter than the condensate to increase the temperature of the condensate. In some embodiments, the fluid at the bottom of the HES **250** is directed to the solar tower (such as the solar tower **110** described in relation to FIG. 1) in order to increase the temperature of the fluid using the solar thermal energy.

[0034] In some embodiments, the heat pump **254**, namely the compressor, is powered using electricity from the energy module, e.g., the electricity provided by the PV panels and/or the heat engine. In some embodiments, the hot fluid from the HES **250** may be used to provide heat to additional units of the CO₂ module, such as for instance the impurity removal/gas polishing unit or gas conditioning unit (not shown on the FIG. 2 drawing).

[0035] In some embodiments, the condensate may be heated to a temperature below the boiling point of the condensate using only a heat exchanger **252** and/or a heat pump **254** against the hot fluid. In some examples, the temperature below the boiling point of the condensate may be sufficient for regeneration.

[0036] In some embodiments, the hot fluid from the HES **250** may be heated (for instance via resistive heating) before the heat exchanger **252** so that the condensate is above boiling point at the exit of the heat exchanger **252**, which may allow for bypassing the heat pump **254** or for the elimination of the heat pump **254** entirely from the system. The resistive heaters may be powered using the electricity generated by the energy module. In some embodiments, the hot fluid is directed to the heat pump **254** and/or heat exchanger **252** before reaching the HES **250**. In some embodiments, another source of heat is available, such as flue gas, and such source of heat is used to heat the condensate alone or in combination with the heat exchanger **252** and/or heat pump **254** described herein.

[0037] FIG. 3 shows a different embodiment of a CO₂ installation **300** according to the present disclosure, used for a different type of CO₂ module **302**. The CO₂ installation **300** includes a CO₂ module **302** configured as a continuous adsorption/desorption system. In some embodiments, the CO₂ module **302** has a vessel **304** including of one or more packed beds containing a capture material in a solid form. In some examples, the capture material is a sorbent. In some examples, the sorbent is mesoporous carbon. The CO₂ module **302** may include, as is shown on FIG. 3, several vessels **304A**, **304B**. The CO₂ module **302** also includes a fluid circuit having one or more fluid distribution devices to fluidly connect each of the vessel **304A**, **304B** to a flue gas line **306** and a clean gas line **307** for adsorption in a first configuration and to a steam line **308** and a loaded steam line **309** for desorption in a second configuration.

[0038] During a first period of time, the flue gas is circulated in the flue gas line **306** to the vessel **304A** (that forms the adsorption/capture unit) in which CO₂ is adsorbed by the capture material

therein, and the clean gas is then discharged to the clean gas line **307**. During the first period of time, the steam is circulated to the vessel **304B** (that forms the regeneration/desorption unit) and loads with the CO₂ from the capture material that exits the vessel **304B** forming a loaded steam. [0039] During the second period of time, the steam is circulated to the vessel **304A** (that form the regeneration unit) and the flue gas to the vessel **304B** (forming the adsorption/capture unit). As the steam flowing through the vessel **304B** heats the loaded capture material, the CO₂ loaded in the capture material is unloaded to the steam flowing therethrough. The loaded steam then exits the vessel **304B** via the loaded steam line **309**, and water is condensed in a condenser **310** that it leaves through water line **311** whereas the gaseous phase is discharged into the atmosphere. Lines **308**, **309**, **311** form a closed steam loop.

[0040] In this embodiment, the steam loop includes heat exchangers **312**, **314**. The heat exchangers heat the water condensate obtained from the condenser **310**. The water condensate obtained from the condenser **310** may be approximately 70° C., and the heat exchanger(s) **312**, **314** boil the condensate back into steam. In the first heat exchanger **312**, the hot fluid from the HES **350** of the energy module (such as the HES **250** and/or energy module described in relation to FIG. 2) and the condensate circulate, increasing the temperature of the condensate to approximately 90° C. In the second heat exchanger **314**, the flue gas circulates against the heated condensate and elevates the temperature of the condensate above the boiling point. In some embodiments, the condensate is heat to approximately 120° C.

[0041] As shown on FIG. 4, a similar principle is applied to another embodiment of a CO₂ installation **400** having a non-continuous (i.e., batch) adsorption-desorption process. In a non-continuous process, the condensate does exchange heat against the flue gas and the heat exchanger **314** of FIG. 3. In some embodiments, the heat exchanger **314** is replaced by a heat pump **454** using the hot fluid from the HES **450** to raise temperature of the condensate (such as described in relation to FIG. 2). In some embodiments, the vessels **404A**, **404B** are connected to the heat source (such as the heat pump and/or exchanger) to heat and desorb the CO₂ from the absorbent bed sequentially. For example, a first vessel **404A** is recharged by the application of heat from the HES **450** while a second vessel **404B** is connected to a CO₂-rich flue gas line **406**. Upon complete or nearly complete desorption of the absorbent bed of the first vessel **404A** and/or saturation of the absorbent bed of the second vessel **404B**, the first vessel **404A** and second vessel **404B** may be switched in connection with the heat source. In some embodiments, more than two vessels are used to facilitate maintenance, storage, and/or reduce downtime of the installation **400**.

[0042] Other embodiments as explained in relation to FIG. 2 may be applied to FIG. 3 and/or 4, i.e., using a heat pump instead of a heat exchanger **312**, additionally heating the hot fluid before heat exchanging it with the condensate, etc. The CO₂ module may also include many variations, and the current disclosure is applicable to any CO₂ capture system, regardless of the capture medium or of the particular architecture of said module.

[0043] FIG. 5 is a system diagram of an installation **500** including an energy module similar to that described in relation to FIG. 1, and the installation **500** may include heat-storing solid units **556** (such as refractory bricks). In some embodiments, the heat-storing solid units **556** are heated using the fluid stream that has recovered thermal energy from the solar panels PV modules **508** and/or the solar tower **510** via a heat exchanger. A condensate line **558** of a CO₂ module (such as CO₂ modules **201**, **302**, **402** described in relation to FIG. 2, FIG. 3, and FIG. 4) may circulate through the heat-storing solid units **556** so as to be heated and/or pre-heated via a heat exchanger to a temperature of approximately 90° C. In another embodiment, the heat-storing solid units may be disposed in the hot fluid loop and/or in the condensate line and may replace and/or supplement other heating devices in the fluid loop (for instance, instead of the heat exchanger **252** in FIG. 2). In some embodiments, the heat-storing solid units are combined with any other additional heating element. In some embodiments, the installation includes a heat exchanger and/or heat pump to exchange heat between the hot fluid and the condensate stream in addition to the heat storing solid

units.

[0044] In at least one embodiment, the solar tower includes two solar receivers, one as explained in relation to FIG. 1 that recovers electricity and heat, and a second that is configured to recover only heat. In such an embodiment, all solar energy converted by the second solar receiver is converted to thermal energy and used to produce a hot fluid stream at a higher temperature than the 90° C. obtained using the system described in FIG. 1. In some embodiments, the hot fluid stream is used to raise the temperature of the condensate stream above its boiling point without using a heat pump, but rather through passive heat exchange.

[0045] In an embodiment, when the installation includes passive heat exchanger (via direct heat exchanger or indirect heat exchange using heat storing solid units) and a heat pump, the heat pump may be powered only in case the passive heat exchange is not able to bring the condensate stream at the right temperature. In that case, the installation may include a controller implementing control schemes for powering the heat pump based on a temperature sensor reading obtained in the condensate line, at the outlet of the heat exchanger. Such control may be beneficial to supply the required heat during the night, when no solar thermal energy is recovered.

[0046] In some instances, the electricity and heat generated by the energy module in real-time are sufficient to heat the regeneration stream and power the refrigeration unit, so that the heat engine is not running. This is, for instance, the case when the sun is shining and is close to the zenith.

[0047] In some instances, electricity and heat generated by the energy module are insufficient to power the heating units and the refrigeration unit, for instance in the night. However, it may be necessary that the CO₂ capture installation continues to operate in such instances. In such a case, the thermal cycle generator (such as the thermal cycle generator 102 described in relation to FIG. 1) may operate using the thermal energy stored in the HES (and/or CES, as described above), so that additional heating units may be used to heat the regeneration stream in replacement or in addition to the heating units used when solar energy is available to the installation. The thermal cycle generator may be triggered by a sensor reading, in particular a temperature of the regeneration stream (i.e., condensate or steam) at the inlet of the regeneration unit and/or the solar energy received by the solar receiver.

[0048] FIG. 6 is a system diagram of an embodiment of an energy module of a CO₂ capture installation 600. In some embodiments, the energy module is part of a direct air capture (DAC) installation. It should be understood that the embodiment of an energy module described in relation to FIG. 6 may be used with non-DAC installations, as well. In some embodiments, a DAC installation 600 according to the present disclosure includes a solar tower 610 that includes a first solar receiver 608 including a PV module that converts a portion of the solar energy to electricity and a thermal receiver that collects solar thermal energy in a fluid loop of a first fluid 661, ie hot fluid, to provide approximately 90° C. fluid to the HES 612, such as described in relation to FIG. 1. In some embodiments, the energy module of the DAC installation 600 further includes a refrigeration unit 614 configured to charge a CES 616. In some embodiments, the HES 612 and CES 616 provide the temperature differential to operate a thermal cycle generator 602. In some embodiments, the installation 600 lacks a CES 616 and/or refrigeration unit 614, and the thermal cycle generator 602 operates on a temperature differential between a heat source (such as the first fluid 661 and/or second fluid 662, as will be described herein) and ambient temperature. The thermal cycle generator 602 and/or the PV module 608 may export electrical power to a grid 618 or a local electrical storage device, such as batteries. In some embodiments, the refrigeration unit 614 is configured to receive electricity from the grid 618 or local electrical storage device.

[0049] In some embodiments, the DAC installation 600 further includes a second solar receiver including a thermal receiver 660 in the solar tower 610 (and/or, optionally, a second thermal receiver in a second solar tower) that raises a temperature of a high-temperature second fluid 662 to greater than 90° C., such as greater than 100° C. or up to approximately 150° C. In some embodiments, the high-temperature second fluid 662 may be pressurized water, steam, oil, or other

gaseous or liquid fluid. In some embodiments, the high-temperature second fluid **662** is in a temperature range of 120° C. and 150° C. In some embodiments, the hot first fluid **661** and the high-temperature second fluid **662** are the same fluid in separate closed loops. In some embodiments, the hot first fluid **661** and the high-temperature second fluid **662** are different fluids. For example, the first fluid **661** may be water, and the second fluid **662** may be oil. In some embodiments, the first fluid **661** and the high-temperature second fluid **662** are the same fluid and intermix in at least one location in the installation **600**. For example, a single shared fluid may be heated to approximately 90° C. in the first solar receiver **608**, and a first portion of the shared fluid is diverted to the HES **612** as a hot first fluid **661** while a second portion of the shared fluid is diverted to toward the second solar receiver including the thermal receiver **660**, which heats the second portion into the high-temperature second fluid **662** with a temperature greater than 100° C. that circulates in another part of the installation **600**. The system may include to this effect a fluid distribution device and a controller to control the flow of the fluid exiting the first solar receiver **608** to the HES and the second solar receiver **660** respectively. Such distribution may be based on parameters of the CO2 module **601**, and/or of the HES (thermocline level for instance).

[0050] As explained above, the HES and/or HTES have a temperature that vary between the top and the bottom of the pit. The temperature at the top of the pit is higher than the temperature at the bottom. In the embodiment of FIG. **6**, a pipe connects the bottom of the HES **612** to the first solar receiver so that fluid from the HES recirculates through the first solar receiver to be heated by the first and/or second solar receiver. A pipe connects the bottom of the HTES to the first and/or second solar receivers so that fluid from the HTES recirculates through the first and/or second solar receiver to be heated by the first solar receiver.

[0051] In some embodiments, a fluid entering the thermal receiver **660** has a temperature less than 100° C., and the high-temperature fluid **662** exiting the thermal receiver **660** has a temperature greater than 100° C. In some embodiments, a fluid entering the thermal receiver **660** has a temperature less than 120° C., and the high-temperature fluid **662** exiting the thermal receiver **660** has a temperature greater than 120° C. In some embodiments, the installation **600** includes a first thermal receiver and a second thermal receiver in series (which, as described herein, may be on a separate solar tower). In such an example, high-temperature fluid **662** exiting the first thermal receiver has a temperature less than 120° C., and the high-temperature fluid **662** exiting the second thermal receiver has a temperature greater than 120° C.

[0052] In some embodiments, at least a portion of the high-temperature fluid **662** is distributed from the thermal receiver **660** to the HES **612**. In some embodiments, at least a portion of the high-temperature second fluid **662** is distributed from the thermal receiver **660** to a CO2 module **601** (as will be described in more detail in relation to FIGS. **7** and **8**). In some embodiments, least a portion of the high-temperature second fluid **662** is distributed from the thermal receiver **660** to a high-temperature energy storage (HTES) **663**. In some embodiments, the HTES **663** is pit storage similar to that described in relation to a HES and/or CES. In some embodiments, the HTES **663** is configured to store the high-temperature fluid (at a temperature higher than the fluid stored in the HES) and provide high-temperature second fluid **662** to the CO2 module **602** and/or to the HES **612** to increase a temperature thereof. The system may include a fluid distribution device and a controller to control the flow of the fluid exiting the thermal receiver **660** to the HES/HTES or CO2 module respectively based on the parameters of the CO2 module and/or the HES/HTES.

[0053] In some embodiment, the HES and/or HTES are configured to provide fluid at the bottom of the pit (having a lower temperature) to circulate back to the solar receivers. For instance, there is a fluid conduit between the bottom of the HES and HTES pits to the first solar receiver **608**.

However, other configurations are possible.

[0054] In some embodiments, the HTES **663** is configured to provide high-temperature second fluid **662** to the thermal cycle generator **602**. For example, the thermodynamic efficiency of the thermal cycle generator **602** may be increased with a higher temperature heat source than the first

fluid **661** of the HES **612**. In particular, a thermal cycle generator **602** operating with an ambient temperature heat sink may benefit from the high-temperature second fluid **662** (either directly from the thermal receiver **660** or from the HTES **663**) as a heat source.

[0055] A DAC installation may be configured to capture CO₂ directly from the atmosphere. Atmospheric CO₂ is much lower concentration than a post-combustion flue gas. For example, ambient atmosphere is approximately 0.2% CO₂, while post-combustion combustion flue gas may be in a range of 2% to 30% CO₂. Energy efficiency is therefore critical to a DAC installation.

[0056] FIG. 7 and FIG. 8 illustrate different applications of the high-temperature second fluid **662** described in relation to FIG. 6 to improve efficiency in a DAC installation. FIG. 7 illustrates an embodiment of a CO₂ module **701** including direct injection of a high-temperature fluid **762**, such as the high-temperature second fluid **662** described in relation to FIG. 6 in which the high-temperature fluid **762** has a temperature greater than 90° C. In some embodiments, the high-temperature fluid **762** has a temperature greater than 100° C. In some embodiments, the high-temperature fluid **762** has a temperature between 100° C. and 150° C.

[0057] The high-temperature fluid **762** is, in some embodiments, directed from a thermal receiver (such as the thermal receiver **660** described in relation to FIG. 6) directly to a first vessel **704A** containing a loaded capture material (adsorbent or absorbent). The high-temperature fluid **762** heats the loaded absorbent, which subsequently desorbs CO₂. A loaded fluid flow (containing steam or other fluid) flows from the first vessel **704A** to a condenser **710**. In some embodiments utilizing steam in the high-temperature fluid **762**, the condenser **710** exhausts heat from the loaded fluid flow, and the condensate and CO₂ flow to a separator or other component that diverts the condensate water and CO₂. In some embodiments, the condensate fluid (e.g. water) is recycled to the solar tower and thermal receiver to be heated again. In some embodiments, the condensate fluid is recycled to an HES.

[0058] In some embodiments, a second vessel **704B** includes an absorbent/adsorbent that captures atmospheric CO₂. In some embodiments, the DAC installation further includes a blower **764** configured to collect and/or compress atmospheric air into a second vessel **704B** containing the absorbent/adsorbent. In some embodiments, the blower **764** is powered by electricity generated by the thermal cycle generator (e.g., ORC generator) such as described in relation to FIG. 1 and FIG. 5. In some embodiments, the blower **764** is powered by electricity generated by the PV modules such as the PV modules **108**, **508** described in relation to FIG. 1 and FIG. 5). As noted herein, the CO₂ concentration in atmospheric air is one to two orders of magnitude less than a post-combustion or industrial flue gas (approximately 0.04% compared to 2%-30% on a weight basis, respectively). In some embodiments, the blower **864** forces an increased flowrate of atmospheric air through the capture unit, thereby increasing amount of CO₂ the absorbent/adsorbent is exposed to and/or contacted with. The blower **864** may, therefore, increase the capture rate of the capture unit. In some embodiments, the blower **864** further compresses the atmospheric air. Compression of the atmospheric air into to the second vessel **704B** and across and/or through the absorbent/adsorbent can increase the mass of CO₂ exposed to the absorbent/adsorbent for the same volume of airflow therethrough. In some embodiments, the installation does not include the blower **764** or **864** and the natural air circulation is used for adsorption.

[0059] FIG. 8 illustrates another embodiment of a CO₂ module **801** including a heat exchanger **866** configured to receive a high-temperature fluid **862**, such as the high-temperature fluid **662** described in relation to FIG. 6. In some embodiments, the high-temperature fluid **862** has a temperature greater than 90° C. In some embodiments, the high-temperature fluid **862** has a temperature greater than 100° C. In some embodiments, the high-temperature fluid **862** has a temperature between 100° C. and 150° C.

[0060] The high-temperature fluid **862** is, in some embodiments, directed from a thermal receiver (such as the thermal receiver **660** described in relation to FIG. 6) to a heat exchanger **866** (which may be similar to the heat exchanger **312** described in relation to FIG. 3). In some embodiments,

the heat exchanger **866** transfers heat from the high-temperature fluid **862** to a closed fluid loop **868** including a working fluid configured to flow into and/or through a first vessel **804A**. In some embodiments, the working fluid is water. In some examples, the heat exchanger **866** vaporizes the water into steam that is circulated through the first vessel **804A**. In some embodiments, the working fluid is water that does not vaporize and remains liquid water through the closed fluid loop **868**. In some embodiments, the closed fluid loop **868** further includes a heater **870** configured to heat the working fluid above a temperature of the high-temperature fluid **862**. For example, the high-temperature fluid **862** may have a temperature less than the boiling temperature of the working fluid, and the heater **870** may heat and/or vaporize the working fluid prior to injection into the first vessel **804A**. In some embodiments, the working fluid is not water. For example, the working fluid may be an oil.

[0061] In some embodiments, the working fluid in the closed fluid loop **868** heats the loaded absorbent in the first vessel **804A**, which subsequently desorbs CO₂. A loaded fluid flow (containing steam or other working fluid) flows from the first vessel **804A** to a condenser **810**. In some embodiments utilizing steam in as the working fluid, the condenser **810** exhausts heat from the loaded fluid flow, and the condensed working fluid and CO₂ flow to a separator or other component that diverts the condensate water and CO₂. In some embodiments, the condensed working fluid (e.g. water) is recycled to the heat exchanger **866** through the closed fluid loop **868**, while the cooled fluid **862** is recycled to the thermal receiver to be heated again. In some embodiments, the cooled fluid **862** is recycled to an HES.

[0062] Similar to the embodiment described in relation to FIG. 7, in some embodiments, a second vessel **804B** includes an absorbent/adsorbent that captures atmospheric CO₂. In some embodiments, the DAC installation further includes a blower **864** configured to collect and/or compress atmospheric air into a second vessel **804B** containing the absorbent/adsorbent. In some embodiments, the blower **764** is powered by electricity generated by the thermal cycle generator (e.g., ORC generator) such as described in relation to FIG. 1 and FIG. 5. In some embodiments, the blower **764** is powered by electricity generated by the PV modules such as the PV modules **108**, **508** described in relation to FIG. 1 and FIG. 5). As noted herein, the CO₂ concentration in atmospheric air is one to two orders of magnitude less than a post-combustion flue gas (approximately 0.4% compared to 2%-30% on a weight basis, respectively). In some embodiments, the blower **864** forces an increased flowrate of atmospheric air through the capture unit, thereby increasing amount of CO₂ the absorbent/adsorbent is exposed to and/or contacted with. The blower **864** may, therefore, increase the capture rate of the capture unit. In some embodiments, the blower **864** further compresses the atmospheric air. Compression of the atmospheric air into to the second vessel **804B** and across and/or through the absorbent/adsorbent can increase the mass of CO₂ exposed to the absorbent/adsorbent for the same volume of airflow therethrough.

[0063] FIG. 9 is a flowchart illustrating an embodiment of a method **972** of capturing CO₂. In some embodiments, the method **972** includes producing electrical power and heat from solar energy using an energy module at **974**. In some embodiments, the energy module is any embodiment of an energy module described herein, such as in relation to FIG. 1. In some embodiments, the energy module includes a solar tower configured to convert at least a first part of the solar energy into solar electrical power via one or more PV modules and at least a second part of the solar energy into solar thermal power (e.g., heat) via a fluid circulating through the solar tower. In some embodiments, the energy module includes a first solar receiver or tower configured to convert at least a first part of the solar energy into solar electrical power via one or more PV modules and a second solar tower or receiver configured to convert at least a second part of the solar energy into solar thermal power (e.g., heat) via a fluid circulating through the second solar tower. In some embodiments, the first solar receiver or tower is configured to convert the solar energy into solar electrical power and into solar thermal power while the second solar receiver is configured to convert the solar energy into solar thermal power only. A higher amount of solar

energy is therefore converted into solar thermal power in the second receiver than in the first one. In some embodiments, the energy module further includes a thermal cycle generator configured to use at least a portion of the solar heat to produce electrical power. In some embodiments, the energy module further includes an HES, as described herein. In some embodiments, the energy module further includes a CES, as described herein.

[0064] In some embodiments, the method **972** further includes capturing CO₂ from a gas using a capture material at **976**. In some embodiments, the capture material is an absorbent. In some embodiments, the capture material is an adsorbent. In some embodiments, the capture material is or includes carbon capture solvents, amines, and sorbent materials (metal-organic frameworks, Mesoporous carbon, etc.) In some embodiments, the capture material is a solid. In some embodiments, the capture material is a liquid.

[0065] In some embodiments, the gas is atmospheric air. In some embodiments, capturing the CO₂ includes compressing the atmospheric air prior to exposing the capture material to the atmospheric air. In some embodiments, the gas is post-combustion flue gas. In some embodiments, capturing the CO₂ includes compressing the post-combustion flue gas prior to exposing the capture material to the post-combustion flue gas. In some embodiments, capturing the CO₂ includes cooling the post-combustion flue gas prior to exposing the capture material to the post-combustion flue gas. In some embodiments, the gas is natural gas. In some embodiments, capturing the CO₂ includes compressing the natural gas prior to exposing the capture material to the natural gas. In some embodiments, capturing the CO₂ includes cooling the natural gas prior to exposing the capture material to the natural gas.

[0066] In some embodiments, the method **972** further includes regenerating the loaded capture material, including unloading the CO₂ from the loaded capture material, using heat, wherein at least a portion of the heat produced by the energy module is used in regenerating the loaded capture material at **978**. In some embodiments, the heat is provided from an HES of the energy module. In some embodiments, the heat is provided by a hot working fluid of the HES transferring heat to the loaded capture material. In some embodiments, the heat is provided to a closed fluid loop (e.g., a regeneration stream, as described herein) via a heat exchanger in thermal communication with the HES and/or hot fluid thereof. In some embodiments, the heat is provided directly from a thermal receiver of a solar tower of the energy module. In some embodiments, the heat is provided a liquid working fluid (e.g., water). In some embodiments, the heat is provided a gaseous working fluid (e.g., steam). In some embodiments, the working fluid that regenerates the capture material has a temperature greater than 90° C. when transferring heat to the capture material. In some embodiments, the working fluid that regenerates the capture material has a temperature greater than 100° C. when transferring heat to the capture material. In some embodiments, the working fluid that regenerates the capture material has a temperature greater than 120° C. when transferring heat to the capture material.

[0067] The present disclosure relates generally to a combined carbon capture and electrical power generation installation and methods of operation thereof according to at least the following clauses:

[0068] Clause 1. A CO₂ capture installation, comprising: a CO₂ capture module, including a capture unit for capturing CO₂ from a gas using a capture material and a regeneration unit for unloading the CO₂ from the loaded capture material and regenerating said capture material using heat; and an energy module for producing electrical power and heat using solar energy, wherein at least a portion of the heat produced in the energy module is used in the regeneration unit.

[0069] Clause 2. The CO₂ capture installation according to the preceding clause, wherein the heat produced in the energy module is used to heat a regeneration stream to be circulated in the regeneration unit to a temperature greater than 90° C., optionally 105° C.

[0070] Clause 3. The CO₂ capture installation according to any preceding clause, wherein the energy module includes at least one solar receiver for receiving solar rays, a hot fluid circulating in the solar receiver to recover thermal energy from the solar rays and a storage pit, optionally

thermally insulated, for storing the hot fluid.

[0071] Clause 4. The CO₂ capture installation according to the preceding clause, wherein the solar receiver is mounted on a solar tower and wherein the energy module includes one or more mirrors, optionally having an adjustable orientation, to direct the solar rays to the solar receivers.

[0072] Clause 5. The CO₂ capture installation according to clause 3 or 4, comprising one or more heat exchangers, such as a direct or indirect heat exchanger, for exchanging heat between the hot fluid and the regeneration stream.

[0073] Clause 6. The CO₂ capture installation according to the preceding clause, wherein the heat exchanger includes a heat storing installation comprising solid heat storing units, such as refractory bricks, configured so that the hot fluid and the regeneration stream both exchange heat with the solid heat storing elements.

[0074] Clause 7. The CO₂ capture installation according to the preceding clause, comprising a hot fluid line from the solar receiver to the storage pit wherein the heat storing installation is upstream from the storage pit in the hot fluid line.

[0075] Clause 8. The CO₂ capture installation according to any of clauses 5-7, including a hot fluid loop to fluidly connect the storage pit to at least one of the heat exchangers.

[0076] Clause 9. The CO₂ capture installation according to any of clauses 2-8, including a heat pump wherein the hot fluid circulates in an evaporator of the heat pump, and the regenerating stream circulates in a condenser of said heat pump.

[0077] Clause 10. The CO₂ capture installation of the preceding clause, including a hot fluid loop to fluidly connect the storage pit to the heat pump.

[0078] Clause 11. The CO₂ capture installation according to clauses 9-10, comprising a regeneration stream line for carrying the regeneration stream to the regeneration unit, wherein the heat exchanger is situated in the regeneration stream line upstream from the condenser of the heat pump, so that the regeneration stream at the outlet of the heat exchanger has a first temperature and has a second temperature higher than the first temperature, and optionally higher than its boiling point, at the outlet of the heat pump.

[0079] Clause 12. The CO₂ capture installation of the preceding clause, including a controller for powering the heat pump based on a temperature of the regeneration stream at the outlet of the heat exchanger.

[0080] Clause 13. The CO₂ capture installation of any preceding clause, comprising one or more additional heating unit not using the heat produced by the energy module, optionally including one or more additional heat exchangers, additional heat pumps, resistive heating elements, etc.

[0081] Clause 14. The CO₂ capture installation according to any preceding clause, wherein at least one of the solar receivers of the energy module is a photovoltaic solar panel and generates electricity.

[0082] Clause 15. The CO₂ capture installation of the preceding clause, wherein the electricity generated by the photovoltaic solar panel is used to power the heat pump and/or an additional heating element and/or an element of the CO₂ module, such as a fluid circulation pump, a valve.

[0083] Clause 16. The CO₂ capture installation of any preceding clause, wherein the storage pit is a hot storage pit, wherein the energy module further includes a refrigeration unit to cool a cold fluid and a cold storage pit, optionally thermally insulated, to store the cold fluid.

[0084] Clause 17. The CO₂ capture installation of the preceding clause, wherein electricity generated by a photovoltaic solar panel of the energy module is used to power the refrigeration unit.

[0085] Clause 18. The CO₂ capture installation of the preceding clause, wherein the hot fluid and cold fluid are used in a heat engine, such as an ORC generator or a Kalina generator, to generate electricity.

[0086] Clause 19. The CO₂ capture installation of any preceding clause, wherein the energy module includes a first solar receiver that produces electricity and heat and a second solar receiver

that produces heat, wherein a first hot fluid circuit is configured so that a first hot fluid recovers thermal energy from the first solar receiver and a second hot fluid circuit is configured so that a second hot fluid recovers thermal energy from the second receiver.

[0087] Clause 20. The CO₂ capture installation of the preceding clause, wherein the second hot fluid exchanges heat with the regeneration stream and the first hot fluid is stored in the hot storage pit without exchanging thermal energy with the regeneration stream for use in the heat engine.

[0088] Clause 21. The CO₂ capture installation of any preceding clause, wherein the capture material is a solvent, a sorbent, or an ion exchange unit, or combination thereof.

[0089] Clause 22. The CO₂ capture installation of any preceding clause, wherein the gas is atmospheric air, post-combustion flue gas, a natural gas.

[0090] Clause 23. The CO₂ capture installation of any preceding clause, wherein the regeneration stream is steam.

[0091] Clause 24. The CO₂ capture installation of any preceding clause, wherein the capture unit includes one or more packed beds, a rotating packed bed, an absorption column, a structured sorbent for circulating the gas counter current with the capture material.

[0092] Clause 25. A method for CO₂ capture comprising: producing electrical power and heat from solar energy using an energy module, capturing CO₂ from a gas using a capture material, regenerating the loaded capture material, including unloading the CO₂ from the loaded capture material, using heat, wherein at least a portion of the heat produced by the energy module is used in the regenerating of the loaded capture material.

[0093] Clause 26. The method according to any preceding clause, wherein regenerating the loaded capture material includes heating a regeneration stream to a temperature greater than 90° C., optionally 105° C., wherein the regeneration stream is optionally steam.

[0094] Clause 27. The method according to any clause 25-26, wherein producing heat from solar energy includes receiving solar rays at least at one solar receiver, circulating a hot fluid in the solar receiver to recover thermal energy from the solar rays and storing the hot fluid in a hot storage pit, optionally thermally insulated.

[0095] Clause 28. The method according to the preceding clause, wherein the solar receiver is mounted on a solar tower and producing heat from solar energy includes directing one or more mirrors receiving the solar rays, optionally having an adjustable orientation, to the at least one solar receiver.

[0096] Clause 29. The method according to clause 27 or 28, comprising exchanging heat between the hot fluid and the regeneration stream via one or more heat exchangers, such as a direct or indirect heat exchanger.

[0097] Clause 30. The method according to the preceding clause, wherein exchanging heat includes storing heat from the hot fluid in a heat storing installation comprising solid heat storing units, such as refractory bricks, and unloading heat from the heat storing installation in the regeneration stream.

[0098] Clause 31. The method according to the preceding clause, wherein storing heat from the hot fluid in the heat storing installation is performed while the hot fluid circulates from the one or more solar receivers to the storage pit.

[0099] Clause 32. The method according to any of clauses 29-31, wherein exchanging heat between the hot fluid and the regeneration stream via at least one heat exchanger includes circulating the hot fluid from the storage pit to the heat exchanger and back to the storage pit.

[0100] Clause 33. The method according to any of clauses 27-32, comprising exchanging heat between the hot fluid and the regeneration stream via one or more heat pump, including circulating the regenerating stream circulates in a condenser of said heat pump and circulating the hot fluid in an evaporator of said heat pump.

[0101] Clause 34. The method of the preceding clause, wherein exchanging heat between the hot fluid and the regeneration stream via at least one heat pump includes circulating the hot fluid from

the storage pit to the heat pump and back to the storage pit.

[0102] Clause 35. The method according to any of clauses 33-34, comprising circulating the regeneration stream to the heat exchanger and then to the condenser of the heat pump, so that the regeneration stream at the outlet of the heat exchanger has a first temperature and has a second temperature higher than the first temperature, and optionally higher than its boiling point, at the outlet of the heat pump.

[0103] Clause 36. The method of the preceding clause, including powering the heat pump based on a temperature of the regeneration stream at the outlet of the heat exchanger.

[0104] Clause 37. The method of any preceding clause, comprising additional heating the regeneration stream without using the heat produced by the energy module, optionally including one or more additional heat exchangers, additional heat pumps, resistive heating elements, etc.

[0105] Clause 38. The method of any preceding clause, wherein producing electricity including using a photovoltaic solar panel from the energy module, wherein the photovoltaic panel optionally also provides heat.

[0106] Clause 39. The method of the preceding clause, including using the electricity generated by the photovoltaic solar panel to power the heat pump and/or an additional heating element and/or an element of the CO₂ module, such as a fluid circulation pump, a valve.

[0107] Clause 40. The method of any preceding clause, wherein the storage pit is a hot storage pit, further including powering a refrigeration unit to cool a cold fluid and storing the cold fluid in a cold storage pit, optionally thermally insulated.

[0108] Clause 41. The method of the preceding clause, wherein the electricity generated by the photovoltaic solar panel is used to power the refrigeration unit.

[0109] Clause 42. The method of the preceding clause 41 or 42, including using the hot fluid and cold fluid in a heat engine, such as an ORC generator or a Kalina generator, to generate electricity.

[0110] Clause 43. The method of any preceding clause, wherein producing electricity and heat using the energy module includes directing solar rays to a first solar receiver that produces electricity and heat and to a second solar receiver that produces heat, circulating a first hot fluid in the first solar receiver and a second hot fluid in the second receiver.

[0111] Clause 44. The method of the preceding clause, including exchanging heat between the second hot fluid exchanges heat and the regeneration stream, and storing the first hot fluid in the hot storage pit without exchanging heat with the regeneration stream, optionally for use in the heat engine.

[0112] Clause 45. The method of any preceding clause, wherein the capture material is a solvent, a sorbent, a ion exchange unit.

[0113] Clause 46. The method of any preceding clause, wherein the gas is atmospheric air, post-combustion flue gas, a natural gas.

[0114] Clause 47. The method of any preceding clause, wherein the gas and the capture unit contact in one or more packed beds, a rotating packed bed, an absorption column, structured sorbent for circulating the gas counter current with the capture material.

[0115] Clause 48. A method for CO₂ capture comprising: producing electrical power and heat from solar energy using an energy module, wherein producing electrical power and heat includes: receiving solar rays at one or more solar receivers, wherein at least one of the solar receiver includes a photovoltaic panel, converting a first portion of the solar energy into electricity including the photovoltaic panel, circulating a hot fluid in the solar receiver to recover a second portion of the energy from the solar rays and storing the hot fluid in a hot storage pit, optionally thermally insulated, and powering a refrigeration unit to cool a cold fluid and storing the cold fluid in a cold storage pit, optionally thermally insulated, the refrigeration unit being optionally powered by the electricity generated by the photovoltaic panel; capturing CO₂ from a gas using a capture material; regenerating the loaded capture material, including unloading the CO₂ from the loaded capture material, using a regeneration stream; and using at least a portion of the heat and/or

electricity produced by the energy module for heating a regeneration stream, including: in a first configuration, exchanging heat with the hot fluid using one or more of a heat exchanger and/or powering one or more first heating units using the electricity produced by the photovoltaic panel, and in a second configuration, using the hot fluid and cold fluid to run a heat engine generating electricity and using the generated electricity to power one or more second heating elements, wherein optionally the second heating units include the first heating unit and additional heating units.

[0116] Clause 49. The method of clause 48, including running the heat engine and using the heat according to the second configuration upon sensing a temperature of the regeneration stream, optionally sensing that the temperature is below a threshold.

[0117] Clause 50. The system or method of any preceding clause, wherein the energy module further includes a first solar tower and a second solar tower, and the first solar tower includes a photovoltaic module and a first thermal receiver and the second solar tower includes a second thermal receiver and has no photovoltaic module.

[0118] Clause 51. The system or method of clause 50, wherein the first thermal receiver is configured to heat a high-temperature fluid to a first temperature greater than 100° C. and the second thermal receiver is configured to receive the high-temperature fluid from the first thermal receiver and heat the high-temperature fluid to a second temperature greater than 120° C.

[0119] Clause 52. A method for CO₂ capture comprising producing electrical power and heat from solar energy using an energy module, wherein producing electrical power and heat includes: receiving solar rays at one or more solar receivers, wherein at least one of the solar receiver includes a photovoltaic panel, converting a first portion of the solar energy into electricity including the photovoltaic panel, circulating a hot fluid in the solar receiver to recover a second portion of the energy from the solar rays and storing the hot fluid in a hot storage pit, and capturing CO₂ from a gas using a capture material to form a loaded capture material; regenerating the loaded capture material, including unloading the CO₂ from the loaded capture material, using a regeneration stream; and using at least a portion of the heat and/or electricity produced by the energy module for heating the regeneration stream, including: in a first configuration, using the hot fluid to heat the regeneration stream, and in a second configuration, using the hot fluid to operate a thermal cycle generator to generate electricity and using the generated electricity to power a heating unit.

[0120] Clause 53. A CO₂ capture installation, comprising a CO₂ capture module, including a capture unit for capturing CO₂ from a gas using a capture material and a regeneration unit for unloading the CO₂ from a loaded capture material and regenerating said loaded capture material using heat; and an energy module for producing electrical power and heat using solar energy, wherein at least a portion of the heat produced in the energy module is used in the regeneration unit.

[0121] Clause 54. The CO₂ capture installation of Clause 53, wherein the heat produced in the energy module is used to heat a regeneration stream to be circulated in the regeneration unit to a temperature greater than 90° C.

[0122] Clause 55. The CO₂ capture installation of Clause 54, wherein the energy module includes a solar receiver for receiving solar rays, a hot fluid circulating in the solar receiver to recover thermal energy from the solar rays and a hot energy storage (HES) for storing the hot fluid.

[0123] Clause 56. The CO₂ capture installation of Clause 55, comprising one or more heat exchangers for exchanging heat between the hot fluid and the regeneration stream.

[0124] Clause 57. The CO₂ capture installation of Clause 56, wherein the heat exchanger includes a heat storing installation comprising solid heat storing units, such as refractory bricks, configured so that the hot fluid and the regeneration stream both exchange heat with the solid heat storing units.

[0125] Clause 58. The CO₂ capture installation of any of Clauses 53-57, including a heat pump

wherein the hot fluid circulates in an evaporator of the heat pump, and the regeneration stream circulates in a condenser of said heat pump.

[0126] Clause 59. The CO₂ capture installation of Clause 58, further comprising one or more heat exchangers for exchanging heat between the hot fluid and the regeneration stream and including a controller for powering the heat pump based on a temperature of the regeneration stream at an outlet of the heat exchanger.

[0127] Clause 60. The CO₂ capture installation of any Clauses 54-59, comprising a regeneration stream line for carrying the regeneration stream to the regeneration unit, wherein a heat exchanger is situated in the regeneration stream line upstream from a condenser of a heat pump, so that the regeneration stream has a first temperature at an outlet of the heat exchanger and has a second temperature higher than the first temperature at an outlet of the heat pump.

[0128] Clause 61. The CO₂ capture installation of any Clauses 53-60, wherein the energy module includes a photovoltaic module and generates solar electrical power.

[0129] Clause 62. The CO₂ capture installation of the preceding Clause, wherein the solar electrical power is used to power at least one element of the CO₂ capture module.

[0130] Clause 63. The CO₂ capture installation of Clause 61 or 62, wherein the energy module further includes a refrigeration unit to cool a cold fluid of a cold energy storage (CES), and the solar electrical power generated by a photovoltaic module of the energy module is used to power the refrigeration unit.

[0131] Clause 64. The CO₂ capture installation of any Clauses 53-63, wherein energy module further includes a thermal cycle generator.

[0132] Clause 65. The CO₂ capture installation of any Clauses 53-64, wherein the energy module further includes a first solar receiver including a photovoltaic module and a thermal receiver configured to heat a fluid at a temperature less than 100° C. and a second solar receiver including one or more thermal receivers configured to heat a high-temperature fluid to a temperature greater than 100° C.

[0133] Clause 66. The CO₂ capture installation of Clause 65, including a high temperature energy storage (HTES) connected to the outlet of at least one of the one or more thermal receivers to store at least a portion of the high temperature fluid.

[0134] Clause 67. The CO₂ capture installation of Clause 65 or 66; having a first thermal receiver configured to heat the high temperature fluid to a first temperature greater than 100C and a second thermal receiver configured to receive the high temperature fluid from the first thermal receiver and heat the high temperature fluid to a second temperature greater than 120C.

[0135] Clause 68. A method for CO₂ capture comprising producing solar electrical power and solar thermal energy from solar energy using an energy module; capturing CO₂ from a gas using a capture material to form a loaded capture material; and regenerating the loaded capture material, including unloading the CO₂ from the loaded capture material using heat, wherein at least a portion of the solar thermal energy produced by the energy module is used in the regenerating of the loaded capture material.

[0136] Clause 69. The method of Clause 68, wherein regenerating the loaded capture material includes heating a regeneration stream to a temperature greater than 90° C. using the portion of the solar thermal energy.

[0137] Clause 70. The method of Clause 68 or 69, wherein producing solar thermal energy from solar energy includes receiving solar rays at least at a solar receiver, circulating a fluid in the solar receiver to recover solar thermal energy from the solar rays and storing at least a portion of the hot fluid in a hot energy storage (HES).

[0138] Clause 71. The method of Clause 70, the at least one solar receiver includes one or more PV modules to generate solar thermal energy and electricity from solar energy and one or more thermal receivers, wherein circulating the fluid in the PV module yields a hot fluid at a temperature of the hot fluid above 80° C. and below 100° C., and wherein circulating the fluid in the one or more

thermal receivers yields a high-temperature fluid at a temperature above **100° C.**

[0139] Clause 72. The method of Clause 71, further comprising storing at least a portion of the high-temperature fluid in a high-temperature energy storage (HTES) and using at least a portion of the solar thermal energy from the HTES to heat the loaded capture material.

[0140] Clause 73. A method for CO₂ capture comprising: producing electrical power and heat from solar energy using an energy module, wherein producing electrical power and heat includes: receiving solar rays at one or more solar receivers, wherein at least one of the solar receiver includes a photovoltaic panel, converting a first portion of the solar energy into electricity including the photovoltaic panel, circulating a hot fluid in the solar receiver to recover a second portion of the energy from the solar rays and storing the hot fluid in a hot storage pit, and capturing CO₂ from a gas using a capture material to form a loaded capture material; regenerating the loaded capture material, including unloading the CO₂ from the loaded capture material, using a regeneration stream; and using at least a portion of the heat and/or electricity produced by the energy module for heating the regeneration stream, including in a first configuration, using the hot fluid to heat the regeneration stream, and in a second configuration, using the hot fluid to operate a thermal cycle generator to generate electricity and using the generated electricity to power a heating unit.

[0141] While embodiments have been described herein, those skilled in the art, having benefit of this disclosure, will appreciate that other embodiments are envisioned that do not depart from the inventive scope. Accordingly, the scope of the present claims or any subsequent claims shall not be unduly limited by the description of the embodiments described herein.

Claims

1. A CO₂ capture installation, comprising: a CO₂ capture module, including a capture unit for capturing CO₂ from a gas using a capture material and a regeneration unit for unloading the CO₂ from a loaded capture material and regenerating said loaded capture material using heat; and an energy module for producing electrical power and heat using solar energy, wherein at least a portion of the heat produced in the energy module is used in the regeneration unit.
2. The CO₂ capture installation of claim 1, wherein the heat produced in the energy module is used to heat a regeneration stream to be circulated in the regeneration unit to a temperature greater than 90° C.
3. The CO₂ capture installation of claim 2, wherein the energy module includes a solar receiver for receiving solar rays, a hot fluid circulating in the solar receiver to recover thermal energy from the solar rays and a hot energy storage (HES) for storing the hot fluid.
4. The CO₂ capture installation of claim 3, comprising one or more heat exchangers for exchanging heat between the hot fluid and the regeneration stream.
5. The CO₂ capture installation of claim 4, wherein the heat exchanger includes a heat storing installation comprising solid heat storing units, such as refractory bricks, configured so that the hot fluid and the regeneration stream both exchange heat with the solid heat storing units.
6. The CO₂ capture installation of claim 2, including a heat pump wherein the hot fluid circulates in an evaporator of the heat pump, and the regeneration stream circulates in a condenser of said heat pump.
7. The CO₂ capture installation of claim 6, further comprising one or more heat exchangers for exchanging heat between the hot fluid and the regeneration stream and including a controller for powering the heat pump based on a temperature of the regeneration stream at an outlet of the heat exchanger.
8. The CO₂ capture installation of claim 2, comprising a regeneration stream line for carrying the regeneration stream to the regeneration unit, wherein a heat exchanger is situated in the regeneration stream line upstream from a condenser of a heat pump, so that the regeneration stream

has a first temperature at an outlet of the heat exchanger and has a second temperature higher than the first temperature at an outlet of the heat pump.

9. The CO₂ capture installation of claim 1, wherein the energy module includes a photovoltaic module and generates solar electrical power.

10. The CO₂ capture installation of claim 9, wherein the solar electrical power is used to power at least one element of the CO₂ capture module.

11. The CO₂ capture installation of claim 9, wherein the energy module further includes a refrigeration unit to cool a cold fluid of a cold energy storage (CES), and the solar electrical power generated by a photovoltaic module of the energy module is used to power the refrigeration unit.

12. The CO₂ capture installation of claim 1, wherein energy module further includes a thermal cycle generator.

13. The CO₂ capture installation of claim 1, wherein the energy module further includes a first solar receiver including a photovoltaic module and a thermal receiver configured to heat a hot fluid at a temperature less than 100° C. and at least a second solar receiver including one or more thermal receivers configured to heat a high-temperature fluid to a temperature greater than 100° C.

14. The CO₂ capture installation of claim 13, including a high temperature energy storage (HTES) connected to the outlet of to at least one of the one or more thermal receivers to store at least a portion of the high temperature fluid.

15. A method for CO₂ capture comprising: producing solar electrical power and solar thermal energy from solar energy using an energy module; capturing CO₂ from a gas using a capture material to form a loaded capture material; and regenerating the loaded capture material, including unloading the CO₂ from the loaded capture material using heat, wherein at least a portion of the solar thermal energy produced by the energy module is used in the regenerating of the loaded capture material.

16. The method of claim 15, wherein regenerating the loaded capture material includes heating a regeneration stream to a temperature greater than 90° C. using the portion of the solar thermal energy.

17. The method of claim 15, wherein producing solar thermal energy from solar energy includes receiving solar rays at least at a solar receiver including a PV module, circulating a fluid in the solar receiver to recover solar thermal energy from the solar rays and storing at least a portion of the hot fluid in a hot energy storage (HES).

18. The method of claim 17, wherein the solar receiver is a first solar receiver, and the fluid circulates from the first solar receiver to a second solar receiver including a thermal receiver configured to further heat the fluid into a high-temperature fluid with a temperature greater than 100° C.

19. The method of claim 18, further comprising storing at least a portion of the high-temperature fluid in a high-temperature energy storage (HTES) and using at least a portion of the solar thermal energy from the HTES to heat the loaded capture material.

20. A method for CO₂ capture comprising: producing electrical power and heat from solar energy using an energy module, wherein producing electrical power and heat includes: receiving solar rays at one or more solar receivers, wherein at least one of the solar receiver includes a photovoltaic panel, converting a first portion of the solar energy into electricity including the photovoltaic panel, circulating a hot fluid in the solar receiver to recover a second portion of the energy from the solar rays and storing the hot fluid in a hot storage pit, and capturing CO₂ from a gas using a capture material to form a loaded capture material; regenerating the loaded capture material, including unloading the CO₂ from the loaded capture material, using a regeneration stream; and using at least a portion of the heat and/or electricity produced by the energy module for heating the regeneration stream, including: in a first configuration, using the hot fluid to heat the regeneration stream, and in a second configuration, using the hot fluid to operate a thermal cycle generator to generate electricity and using the generated electricity to power a heating unit.

